

ONCHOCERCOSE

A. Definition

It's a disease caused by the filarial parasite *Onchocerca volvulus*, which lives under the skin. The major risk is blindness through damage to the cornea and retina.

B. Transmission mode

It is transmitted by the bite of a **simulium** or **INDIRAGUTUMBA**, a small insect close to waterholes; it absorbs the microfilariae by biting a human being and transmits them to another human being.

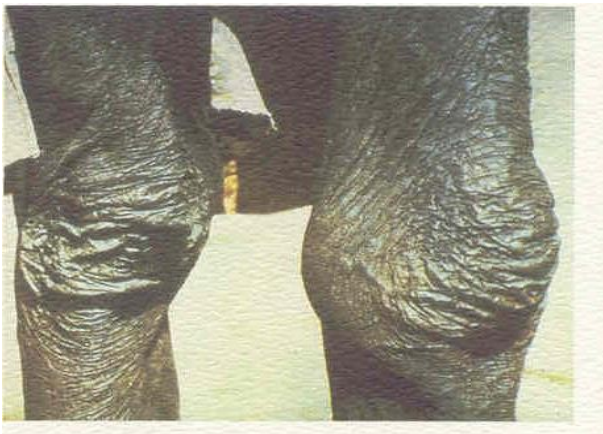
C. Symptomatology:

In many cases, sufferers are unaware of their condition (**asymptomatic onchocerciasis**), only learning of it when symptoms appear by chance.

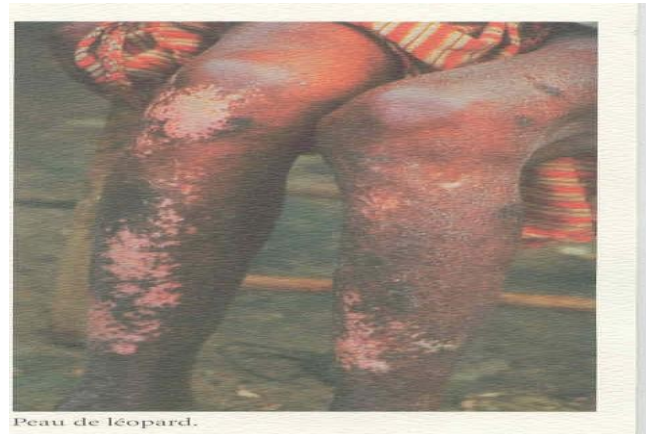
The signs are :

✓ Skin disorders

Its symptoms include intense pruritus, scratching lesions, filarial scabies, thickening of the skin (dermal atrophy with loss of skin elasticity, giving the appearance of ***lizard skin*** and/or depigmentation of the skin on the front of the legs, giving the appearance of ***leopard skin***.

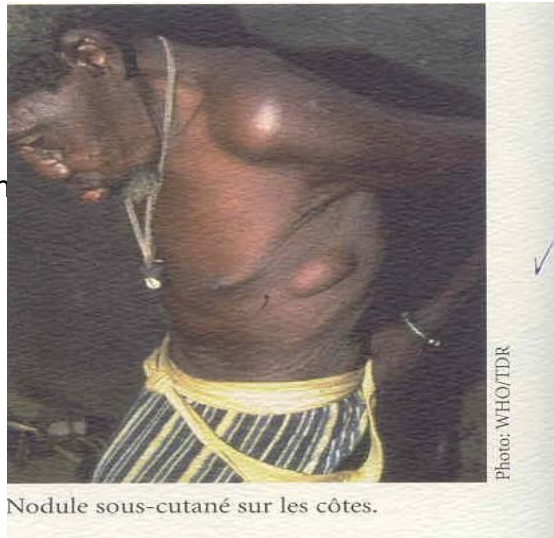


Lizard skin



Leopard skin

Subcutaneous nodules or cysts (hard, mobile, and located under the skin of the ribs,



Multiple nodules on the leg

✓ Lymphatic disorders

Nodes are painless, inguinal swellings suggest a hernia (elephantiasis of the genitals).

✓ Eye signs

Three ocular syndromes that may be isolated or associated:

a) Functional syndrome :

- Night blindness (reduced vision in dim light),
- Reduced visual acuity,
- Visual field narrowing.

b) Anterior segment syndrome:

Immune, linear and sclerosing keratitis, characterized by the presence of whitish spots opacifying the eye, and microfilariae in the anterior chamber of the eye.

c) Posterior segment syndrome characterized by :

- Choroidal and retinal damage (chorioretinal lesions or chorioretinitis)
- Damage to the optic nerve can lead to a narrowing of the visual field.

N.B.: However, the first ocular signs are: tingling eyes, watery eyes, photophobia, progressive loss of visual acuity leading to blindness.

✓ **Other target sites:**

Presence of *Onchocerca volvulus microfilariae* in several organs or biological fluids such blood, brain, liver, kidneys, nerves, CSF, urine, vaginal smears and sputum.

It accompanied by signs such as weight loss and insomnia.

D. Treatment

Ivermectin (Mectizan®) cès 3mg orally

E. Dosage

Oral administration is determined by height and/or weight (150 µg/ Kg) in a single dose.

The following tables show the number of tablets to be administered according to weight or height:

Table I: Mectizan administration according to weight

Body weight (kg)	Dose (in number of 3 mg tablets)
15 à 25	1
26 à 44	2
45 à 64	3
65 and over	4

Table II: Mectizan administration according to size

Size (cm)	Dose (number of 3 mg tablets)
90 à 119	1
120 à 140	2
141 à 159	3
160 more	4

F. Side effects

The most common side effects are :

- **Scratching:** in this case, it should be treated with an anti-histamine.
- **Edema:** the affected person needs an analgesic
- **Fever:** considered serious if it lasts more than 24 hours a day.
In this case, the patient will need to undergo a more thorough examination to identify and treat the real cause of the fever (e.g. malaria).
- **Pain:** if severe, treat with an analgesic.
- **Dizziness or syncope:** these occur fairly rapidly taking Ivermectin. In such cases, patients should be encouraged to rest and drink plenty of fluids.
- **Diarrhea:** if severe, treat with standard rehydration solutions.

G. Contraindications:

Pregnant women, women who have given birth and whose babies are less than 8 days old, children under 5 years of age or weighing less than 15 kg.

N.B. Surgical treatment may be considered for localized nodules.

For isolated clinical cases, ivermectin (MECTIZAN ®) is prescribed at a dose of 200 µg/kg, in a single oral administration, repeated every 6 or 12 months. Ivermectin kills the larvae, but not the adult worms, so annual or semi-annual treatments a long period of time (11 to 14 years) are necessary to the disease under control and prevent any recurrence.